Total Response in AC circuits

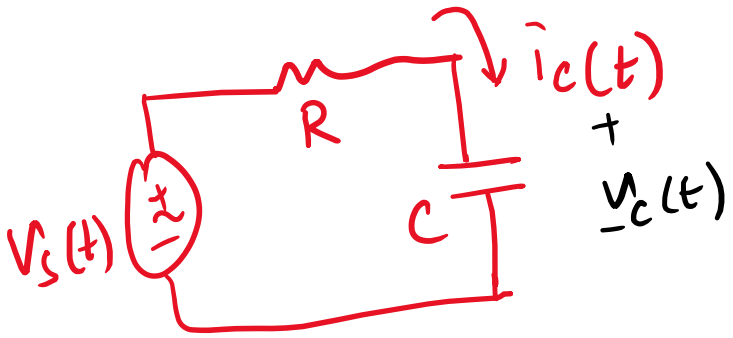
→ Natural Response (stored energy)

+ Forced Response. (Applied energy)

RL, RC circuits

(First order circuits)

RC circuit



KVL : $V_s(t) - i_c(t)R - V_c(t)$

$$i_c(t) = C\frac{dV_c(t)}{dt}$$

$$V_s(t) = RC\frac{dV_c(t)}{dt} + V_c(t)$$

i) Zero input response (natural response)

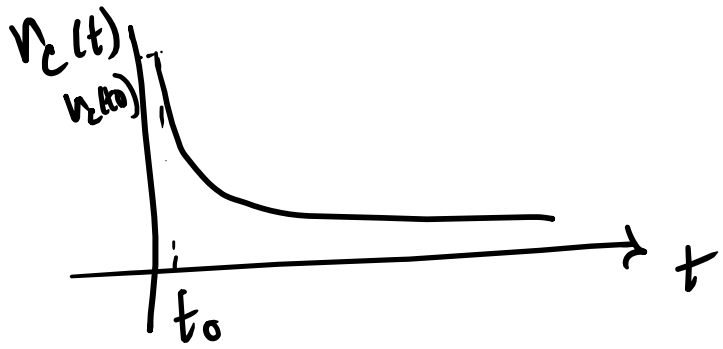
$V_s(t) = 0$

$$\frac{dV_c(t)}{dt} + \frac{1}{RC}V_c(t) = 0$$

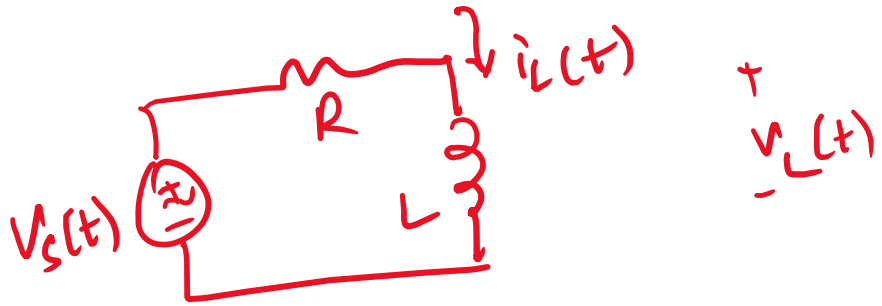
$$\frac{dx(t)}{dt} + \lambda x(t) = 0$$

$$x(t) = x(t_0)\,e^{-\lambda(t-t_0)}$$

$$v_C(t) = v_C(t_0)\, e^{-\frac{1}{RC}(t-t_0)}$$



RL



KVL: $V_s(t) - i_L(t)R - V_L(t) = 0$

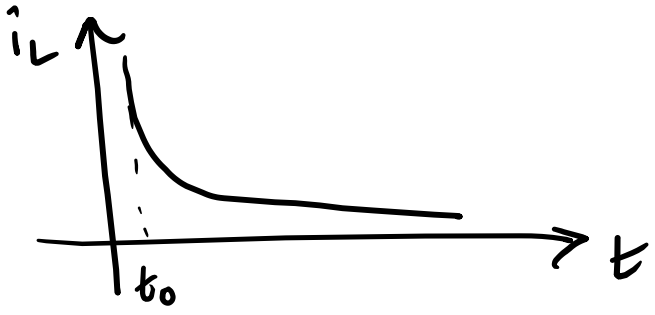
$$V_L(t) = L\frac{di_L(t)}{dt}$$

Zero input response $V_s(t) = 0$.

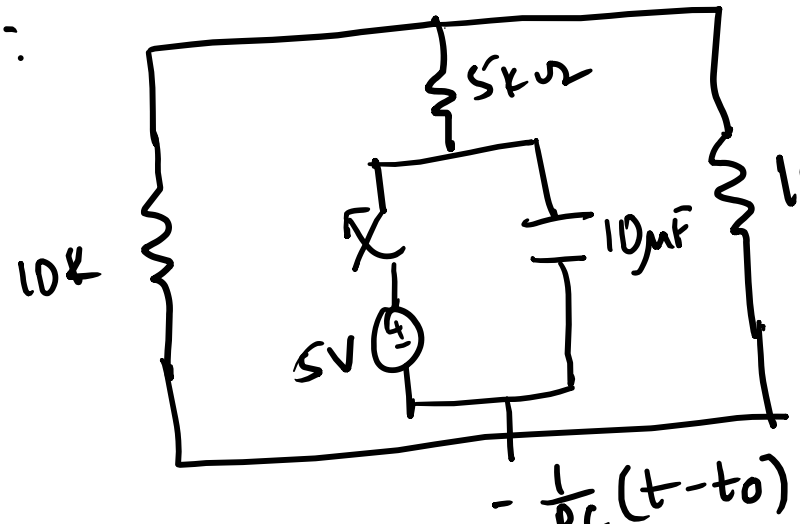
$$R i_L(t) + L\frac{di_L(t)}{dt} = 0$$

$$\frac{di_L(t)}{dt} + \frac{R}{L} i_L(t) = 0$$

$$i_L(t) = i_L(t_0)\, e^{-\frac{R}{L}(t-t_0)}$$



Ex:



Before $t=1s$, switch is closed

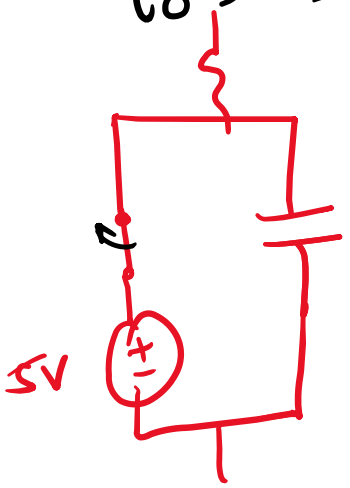
$t \geq 1s$, switch is open.

Find $V_c(t)$ for $t \geq 1s$

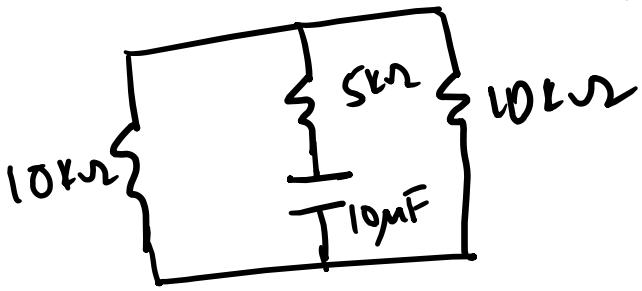
$$V_c(t) = V_c(0)\, e^{-\frac{1}{RC}(t-t_0)}$$

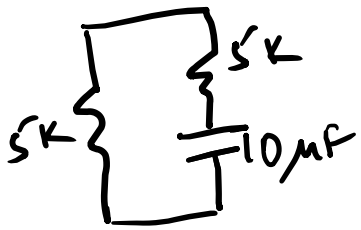
$t_0 = 1s$

$$V_c(1s) = 5V$$

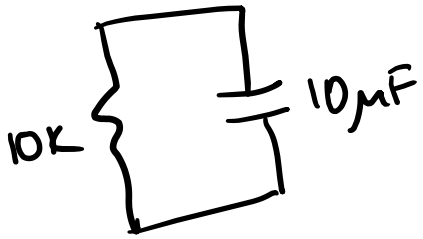


For $t \geq 1s$



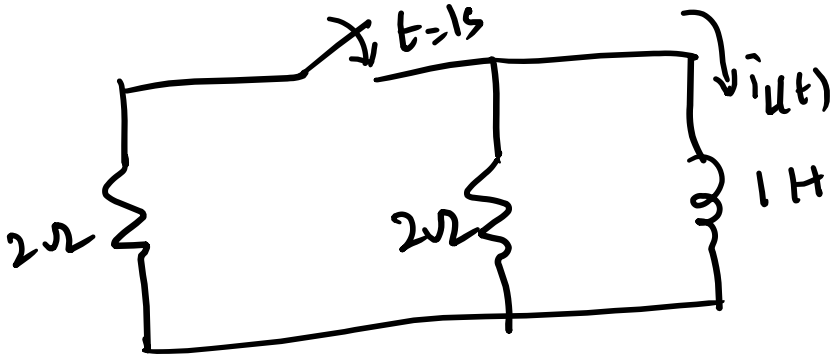


$\rightarrow$



$$V_c(t) = V_c(1) \cdot e^{-\frac{1}{RC}(t-1)}$$

$$V_c(t) = 5\, e^{-10(t-1)}$$



Given

$i_L(0) = 1A$.

Find $i_L(t)$ for $t \geq 0$s

Switch closes at 1s

For $0 \leq t \leq 1$

$$i_L(t) = i_L(0)\, e^{-\frac{R}{L}(t-0)}$$

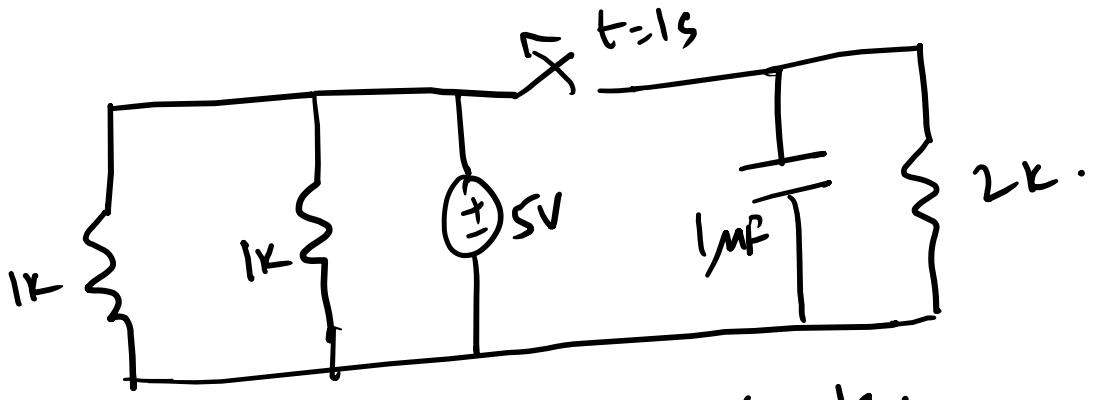
$$\boxed{i_L(t) = 1 \cdot e^{-2t}}$$ $t = 0$ To 1s

For $t \geq 1$ $L = 1H$, $R = 1\Omega \| 2\Omega \{ 2$

$$i_L(t) = i_L(1)\, e^{-(t-1)}$$

$$= 1 \cdot e^{-2t} \cdot e^{-(t-1)}$$

$$\boxed{i_L(t) = e^{-(t+1)}}$$ for $t \geq 1$s



Switch opens at $t = 1s$.

Find $v_c(t)$ for all $t \geq 0s$.

$0 \leq t \leq 1s$ $\quad v_c(t) = 5V$

$t \geq 1s$ $\quad v_c(t) = v_c(1)\, e^{-\frac{1}{RC}(t-1)}$

for $t \geq 1s$ $\quad v_c(t) = 5\, e^{-500(t-1)}$ V

